



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

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SECV2113-HUMAN-COMPUTER INTERACTION

SECTION 1

PROJECT 1

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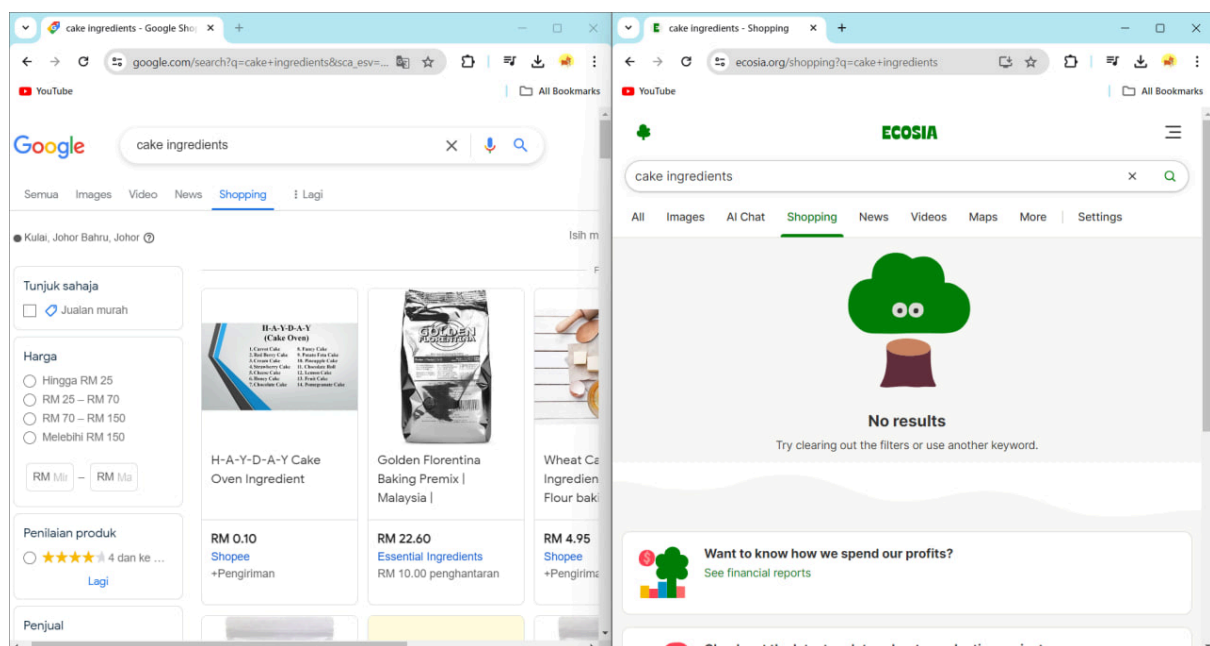
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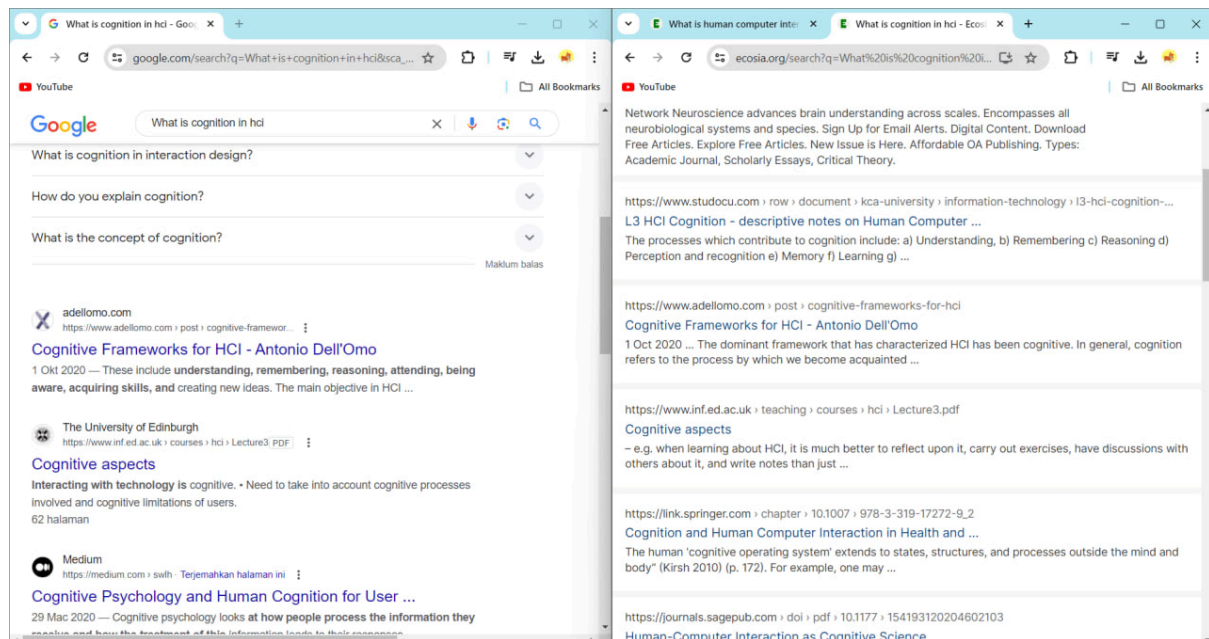
Problem:

In this era of data and information, the primary means for humans to access information is through online resources. Among mainstream search engines like Bing, Google, etc., there is one called Ecosia. The ethos of this search engine is to green the environment and plant trees through advertising revenue, which is also a powerful means to address one of the Sustainable Development Goals, the preservation of life on land. However, despite this noble mission, this search engine still struggles to gain mainstream recognition. Through our thorough usage, we have identified several issues related to user experience.

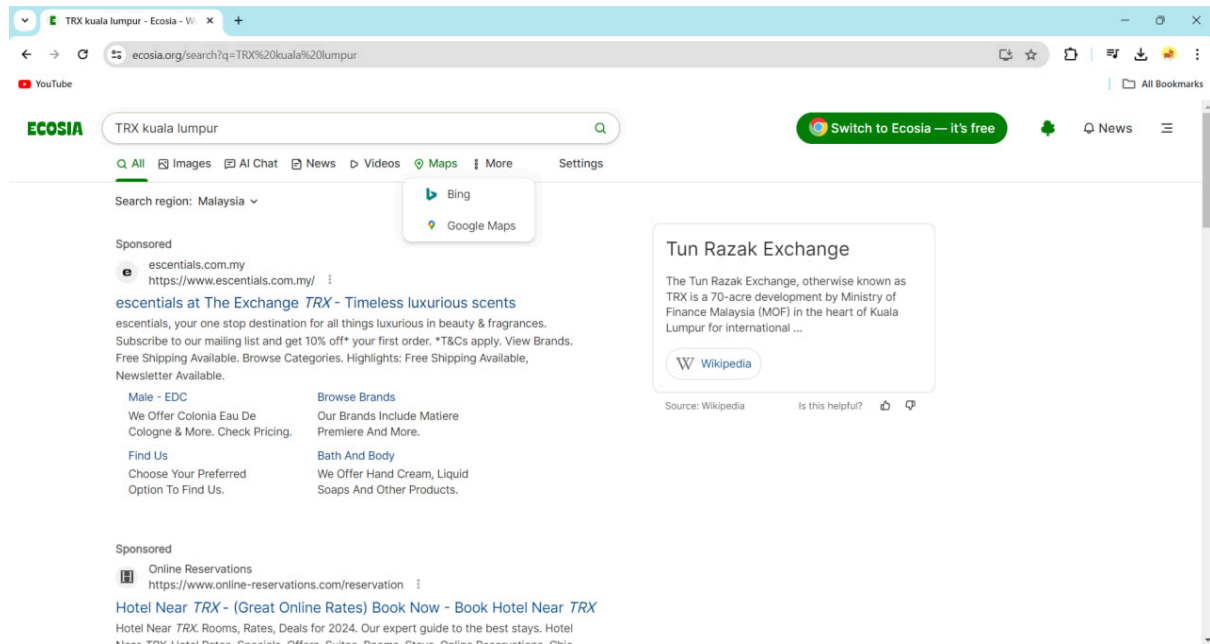
Firstly, Ecosia's search results are not as diverse. When users seek information, a variety of content can enhance their user experience. Currently, Ecosia's search engine contents falls short in comparison to mainstream ones like Google. The lack of content in a search engine is a significant functional issue that greatly diminishes the user experience. Users feeling disappointed when they can't find answers, especially in this era of abundant information, reflects a considerable gap in meeting their expectations. When users have to choose between two search engines, they usually prefer the one that gives them diverse results easily, rather than picking one with a good goal but not enough content. Having a variety of content mainly comes from gathering data and investing in resources, which gives big search engines a big advantage since they've been in this field for a long time. The following images illustrate a comparison between Google's search results for cake-making ingredients and Ecosia's search results for the same. It's evident that Ecosia fails to meet users' expectations regarding content diversity.



Secondly, the layout design is a crucial aspect. Attention and perception in the cognitive process significantly impact user experience. In these aspects, web page layout design should be organised and engaging. However, as seen in the images below, Ecosia's layout design consists solely of text, with similar font sizes throughout, making it become a bad user experience for users to catch the key information. Moreover, Ecosia's layout lacks any visual elements, which can be rather dull. An all-text page can also lead to visual fatigue for users. In contrast, Google's layout design is very comfortable, with clear hierarchies and including some image elements, which greatly enhance user attention and perception.



Thirdly, Ecosia's ecosystem is not comprehensive enough. Some internal applications, such as maps, are not available within Ecosia which is also becoming an usability problem. Whenever users need to use these functionalities, they have to resort to other search engines to complete their tasks. For user experience , they may feel this search engine is very immature due to lack of development. This is undoubtedly a significant flaw because users prefer simplicity. They would rather choose a search engine with a more complete ecosystem from the start than use one that requires assistance from other search engines to accomplish tasks.



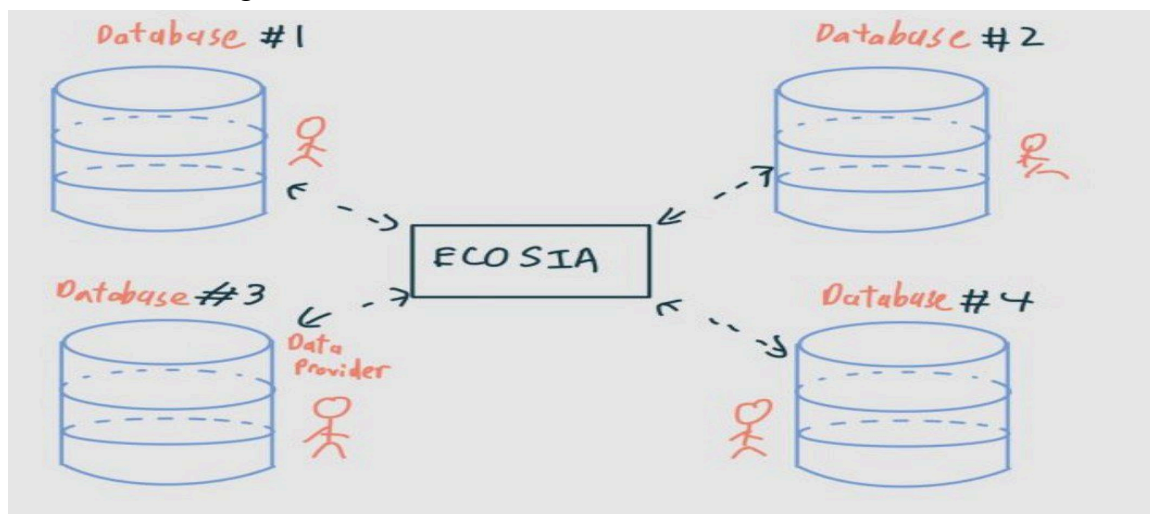
Solution:

If we sought for a better search engine to encourage our users to be invested in tree planting initiatives to combat deforestation, climate change and biodiversity, a few major tweaks and improvements are vital to this search engine to regain its mainstream recognition for its commitment to environmental sustainability. The following are the solutions that we can push out and decide from the problems mentioned above.

First and foremost, to resolve the issue of Ecosia's lack of content diversity and search results, the context of a smart search engine should help with listing relevant information just by typing in first letter or partial names or words.

We can help Ecosia to **collaborate with additional data providers and integrate more sources into its search index**. This could involve partnerships with specialised databases, academic repositories, and content creators to ensure a more comprehensive coverage of information across various domains.

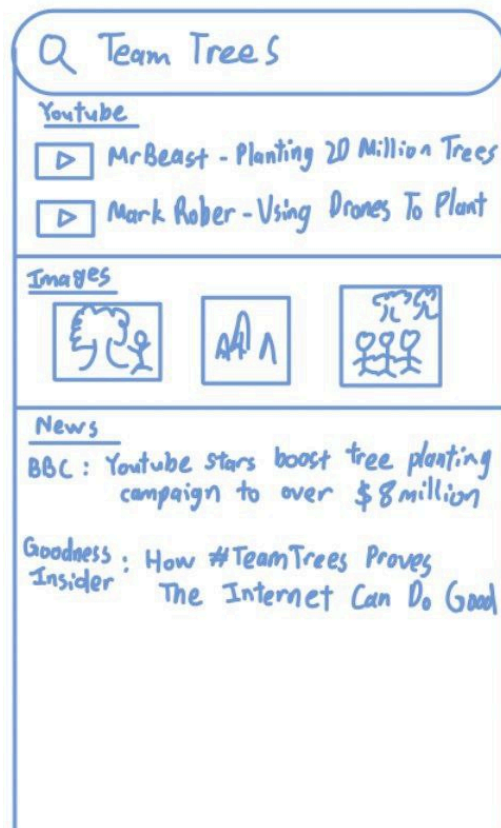
As shown in the figure below:



This method can help Ecosia understand user preferences and improve the relevance and diversity of results over time. By implementing this strategy, thereby enhancing the overall user experience and competitiveness of its search engine in comparison to mainstream alternatives.

Secondly, The other problem that the user faces while using the Ecosia extension is that the layout design is dull due to its layout which just solely contains text which can lead to visual fatigue. We decided to let Ecosia take a different approach such as **implementing a spotlight search tool**.

As shown in the figure below:

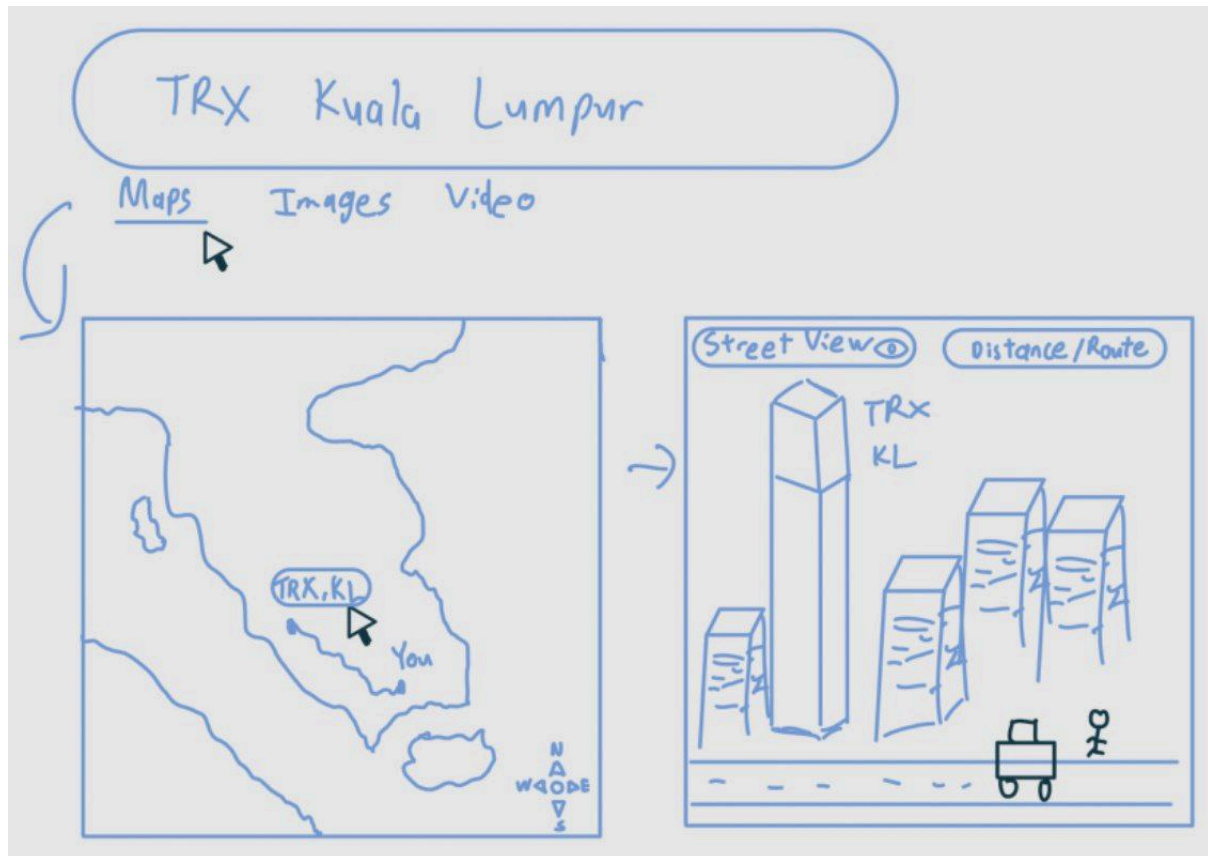


The spotlight search tool can organise search results in a structured layout with clear headings, subheadings, and sections can improve the overall readability and organisation of the page. In addition, features like expandable snippets, quick previews, and filtering options to empower users in their search experience.

This helps users navigate the content more efficiently and locate relevant information with ease.

This solution can lead to the layout design being significantly improved with its search results pages more organised, engaging, and user-friendly. This can contribute to a better overall user experience and increase user satisfaction with the search engine.

Last but not least, We can consider **enhancing Ecosia's functionalities such as in maps by investing in the development of internal mapping capabilities** within the Ecosia platform. As shown in the figure below:



In detail, this can be done through expansion of the features available such as images of the destination, real-time traffic updates, route planning, location-based recommendations, and integration with public transportation information.

With such massive change, this can help Ecosia provide its users with a more robust and user-friendly mapping experience, enhancing the overall functionality and usability of its platform.

Target User

The target users for redesigned Ecosia search engine pages are online researchers that value both functionality and appearance while interacting with websites. This user probably prefers an organised and straightforward layout that improves their ability with easy navigation while also having an eye-catching appearance such as related icon, suitable whitespace, word font, and colour. Additionally, their focus on environmental sustainability would be compatible with Ecosia's objective of planting trees with the help of advertising money. A redesigned Ecosia that enhances the overall design of search result pages, includes the adding of spotlight search tool and unique features like expandable snippets, rapid previews, and filtering choices can provide users who aim to balance their online interactions between sustainability, aesthetics, and functionality users more control over their search experience. that enhances the overall design of search result pages. This will benefit the online researcher by making it easier for them to find pertinent material, navigate the content more quickly, reduce visual fatigue, and improve their entire research experience.

Students are the second target user groups that would benefit significantly from our redesign of Ecosia. Individuals, particularly scholars, are continuously seeking reputable and varying sources of knowledge. They rely extensively on search engines to find data and academic materials for their research. Redesigned Ecosia provides academics with access to a diverse pool of academic materials through collaboration with additional data providers. With more academic content, individuals may study the literature more thoroughly, create better articles, and even publish their own research findings. Furthermore, academic inquiries are continually overflowing with papers and articles that are difficult to draw people's attention and sort through. The searching process can be expedited by using a structured layout, which presents the search results in an intuitive and ordered manner. In this case, Ecosia allows them to swiftly identify crucial sources and critical insights among a vast sea of information.

The third target user groups are environmental enthusiasts and activists. This user group includes individuals and organisations dedicated to environmental concerns such as conservation, climate action, and sustainability advocacy. With enhanced content diversity, they receive access to a diverse range of environmental knowledge, enabling informed discussions, learning new insights, and engaging with diverse perspectives within the environmental community. Moreover, a more engaging layout design promotes a pleasant reading experience, enabling them to explore further into environmental issues. Ecosia's smart search engine helps activists access relevant data and resources. This saves time and effort, allowing activists to acquire critical facts, statistics, and case studies to back up their advocacy work. In addition, the platform's advanced mapping features enable them to visually represent environmental data, plan activities, and cooperate efficiently, resulting in substantial impact on campaigns and projects. Access to real-time information, geographic insights, and location-based recommendations enables activists to make informed decisions and have a positive environmental impact. Hence, the redesign of Ecosia addresses the requirements of environmental enthusiasts and activists, building a community committed to environmental stewardship and action.